

PROJECT II

Modified shocks

1. Euler equations for modified shocks

Under modified shocks, we most commonly refer to shock waves whose physical properties have been altered due to the presence of cosmic radiation. Since cosmic rays are being accelerated at the shocks, the effect of the modification is twofold: the modified Rankine-Hugoniot conditions change the spectrum of cosmic rays, and the cosmic rays themselves, i.e., their pressure, change the conditions at the shocks. The modification primarily manifests upstream of the shock in the so-called precursor; however, we will consider here what influence the presence of cosmic radiation has on the interior of the supernova remnant. This is described by Chevalier's solution (Chevalier 1983), which is a sort of modification of Sedov's (Sedov 1959) and involves the inclusion of cosmic ray pressure in the Euler equations for an ideal fluid.

Let us define, at the beginning, dimensionless quantities: radius $\xi = r/R$, density F , gas pressure G , cosmic ray pressure H , and velocity U :

$$\rho = \frac{\gamma + 1}{\gamma - 1} \rho_0 F(\xi) \quad (1)$$

$$P_g = \frac{2\beta}{\gamma + 1} \rho_0 \dot{R}^2 G(\xi) \quad (2)$$

$$P_{CR} = \frac{2(1 - \beta)}{\gamma + 1} \rho_0 \dot{R}^2 H(\xi) \quad (3)$$

$$v = \frac{2}{\gamma + 1} \dot{R} U(\xi). \quad (4)$$

by following the procedure given in Arbutina (2017), from Euler equations

$$\frac{\partial \rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho v) = 0, \quad (5)$$

$$\rho \frac{Dv}{Dt} = \rho \frac{\partial v}{\partial t} + \rho v \frac{\partial v}{\partial r} = -\frac{\partial P_g}{\partial r} - \frac{\partial P_{CR}}{\partial r}, \quad (6)$$

$$\frac{D(P_g \rho^{-\gamma_g})}{Dt} = \frac{\partial}{\partial t} (P_g \rho^{-\gamma_g}) + v \frac{\partial}{\partial r} (P_g \rho^{-\gamma_g}) = 0, \quad (7)$$

$$\frac{D(P_{CR} \rho^{-\gamma_{CR}})}{Dt} = \frac{\partial}{\partial t} (P_{CR} \rho^{-\gamma_{CR}}) + v \frac{\partial}{\partial r} (P_{CR} \rho^{-\gamma_{CR}}) = 0, \quad (8)$$

where $\frac{D}{Dt}$ is material derivative, we move to their dimensionless equivalents

$$\left(\frac{U}{\xi} - \frac{\gamma+1}{2}\right)\xi\frac{F'}{F} + U' + \frac{2}{\xi}U = 0, \quad (9)$$

$$\xi\left[\left(\frac{2}{\gamma+1}\frac{U}{\xi} - 1\right)U' - \frac{3}{2}\frac{U}{\xi}\right] + \frac{\gamma-1}{\gamma+1}\frac{1}{F}\left[\beta G' + (1-\beta)H'\right] = 0, \quad (10)$$

$$GF^{1-\gamma_g}\left(\frac{U}{\xi} - \frac{\gamma+1}{2}\right) = \frac{1-\gamma}{2\xi^3}, \quad (11)$$

$$HF^{1-\gamma_{CR}}\left(\frac{U}{\xi} - \frac{\gamma+1}{2}\right) = \frac{1-\gamma}{2\xi^3}. \quad (12)$$

In all expressions, we use the notation $f' = \frac{df}{d\xi}$, $f = \{F, G, H, U\}$. We have assumed, of course, that the solution of the Sedov form is given by $R(t) = B(E_o/\rho_o)^{1/5}t^{2/5}$, where E_o is the energy of the explosion and ρ_o is the density of the surrounding (homogeneous) medium.

2. Energy integral

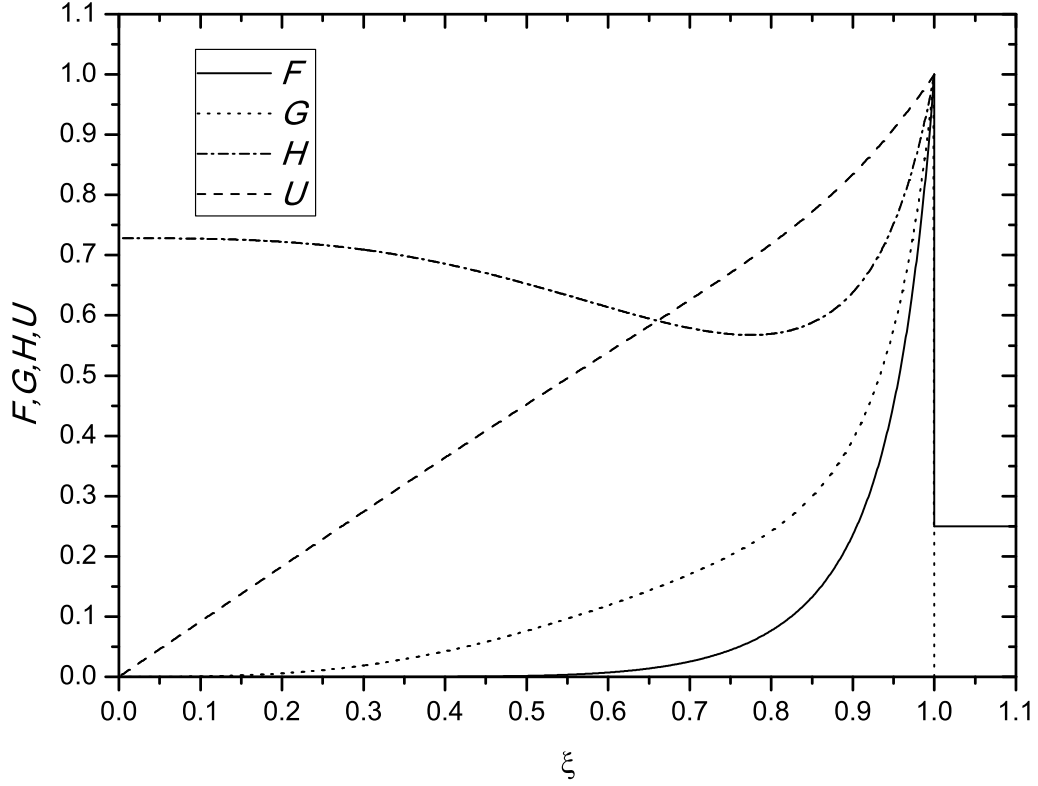
During the Sedov phase in the evolution of the supernova remnant, it is assumed that energy is conserved (negligible radiation losses). If we seek the total energy E_o within the moving sphere defined by the coordinate $\xi = 1$ (shock surface), it must be the sum of the individual components, the kinetic and internal energy of the gas and cosmic rays:

$$\begin{aligned} E_o &= T_o + U_{o,g} + U_{o,CR} \\ &= \int_0^R \left(\frac{1}{2}\rho v^2 + \frac{1}{\gamma_g - 1}P_g + \frac{1}{\gamma_{CR} - 1}P_{CR} \right) 4\pi r^2 dr \\ &= \frac{32\pi E_o}{25(\gamma+1)} B^5 \int_0^1 \left(\frac{1}{\gamma-1}FU^2 + \frac{\beta}{\gamma_g-1}G + \frac{1-\beta}{\gamma_{CR}-1}H \right) \xi^2 d\xi. \end{aligned} \quad (13)$$

The last expression allows us to find the parameter B :

$$B^{-5} = \frac{32\pi}{25(\gamma+1)} \int_0^1 \left(\frac{1}{\gamma-1}FU^2 + \frac{\beta}{\gamma_g-1}G + \frac{1-\beta}{\gamma_{CR}-1}H \right) \xi^2 d\xi, \quad (14)$$

as well as fractions T_o/E_o , $U_{o,g}/E_o$ and $U_{o,CR}/E_o$.



Slika 1: Solution for parameters $\beta = 0.5$, $\gamma_g = 5/3$, $\gamma_{CR} = 4/3$.

3. The problem

Using the dimensionless Euler equations, express $\frac{dU}{d\xi}$ and $\frac{dF}{d\xi}$ as functions of ξ , F , and U , and numerically find the solution using a modified sixth-order Runge-Kutta method (Luther 1968) (see Figure 1). Assume that the shock is significantly modified, i.e., that $\beta = 0.5$, $\gamma_g = 5/3$, $\gamma_{CR} = 4/3$, and effective adiabatic index $\gamma = \frac{5+3(1-\beta)}{3(2-\beta)}$. Calculate the total compression $R = \frac{\gamma+1}{\gamma-1}$, the parameter B , and the percentages of the kinetic and internal energy (of the gas and cosmic rays) in the total energy of the explosion.